

data

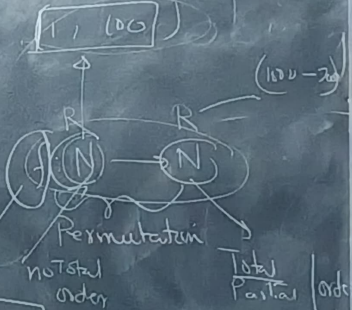
driven approach

System Design

BIAS

Sorting

Problem:



Integers

F.P.

Unknown

Circuit

$f = \{0, 1\}$

$\{0, 1\}$

NAND / AND / INV

$1-100$ { I_1 , I_2 , I_3 }

I

$f \sim f'$

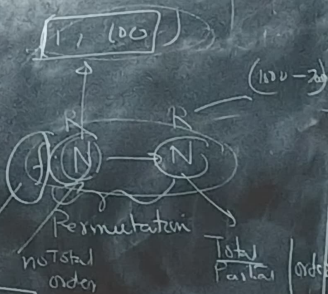
Approximate

BIAS

Sorting

Problem:

Unknown



Circuit

$f: \{0,1\}^n \rightarrow \{0,1\}^m$

VLSI

NAND/AND/INV

Approximate



$I_1, I_2, \dots, I_n$

100

Pulley

M.A

na

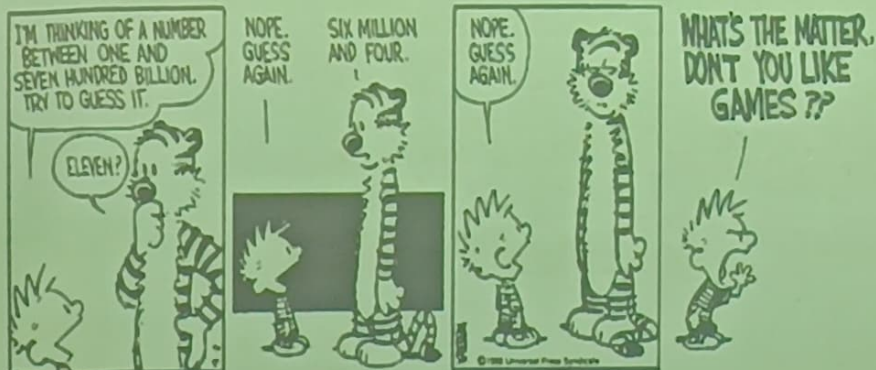
Learning Is Impossible

What's my rule?

- $1\ 2\ 3 \Rightarrow$ satisfies rule
- $4\ 5\ 6 \Rightarrow$ satisfies rule
- $6\ 7\ 8 \Rightarrow$ satisfies rule
- $9\ 2\ 31 \Rightarrow$ does not satisfy rule

Possible rules

- 3 consecutive single digits
- 3 consecutive integers
- 3 numbers in ascending order
- 3 numbers whose sum is less than 25
- 3 numbers < 10
- 1, 4, or 6 in first column
- "yes" to first 3 sequences, "no" to all others



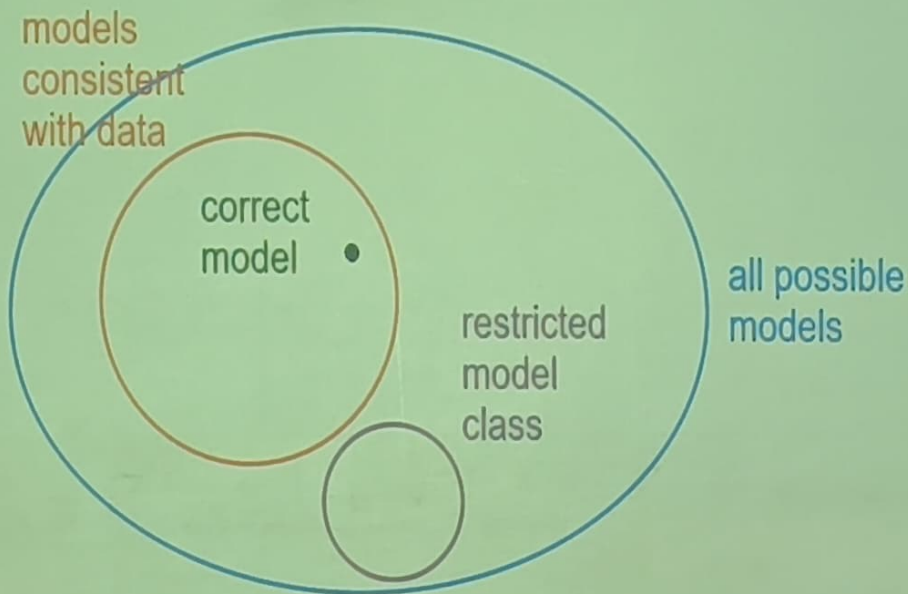
"What's My Rule" For Machine Learning

x1	x2	x3	y
0	0	0	1
0	1	1	0
1	0	0	0
1	1	1	1
0	0	1	?
0	1	0	?
1	0	1	?
1	1	0	?

16 possible rules (models)

With n binary inputs and p training examples, there are $2^{2^n - p}$ possible models.

Model Space



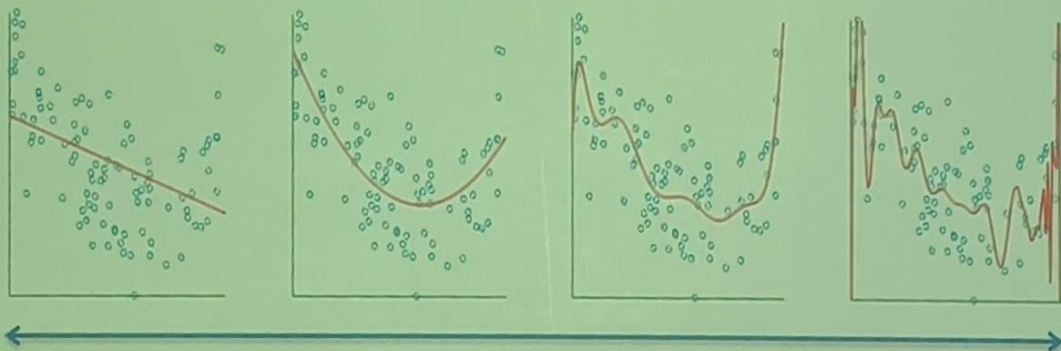
Restricting model class can help

Or it can hurt

Depends on whether restrictions are domain appropriate

Restricting Models

Models range in their flexibility to fit arbitrary data



simple model

complex model

constrained

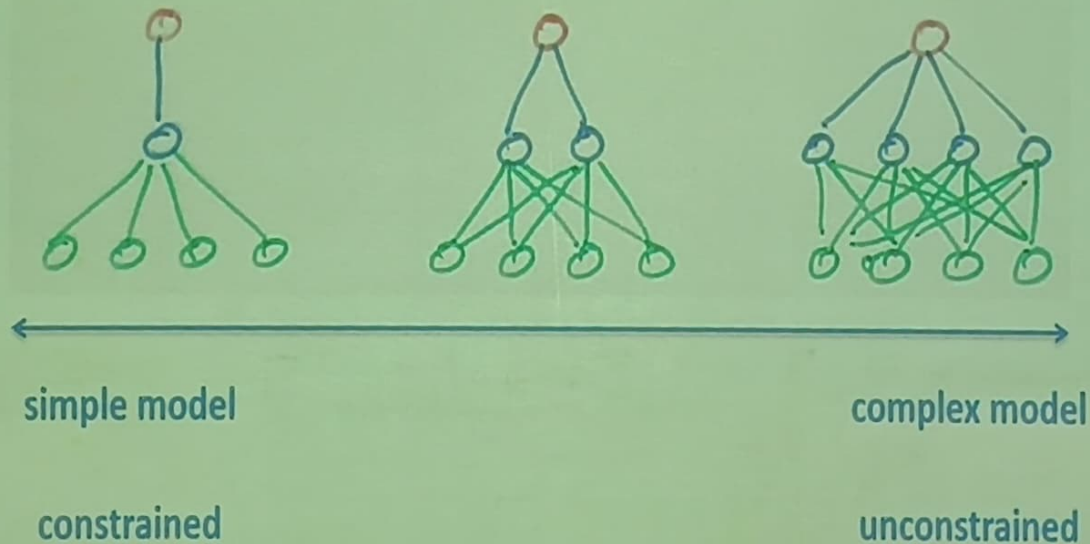
unconstrained

small capacity may
prevent it from
representing all
structure in data

large capacity may
allow it to fit quirks
in data and fail to
capture regularities

Restricting Models

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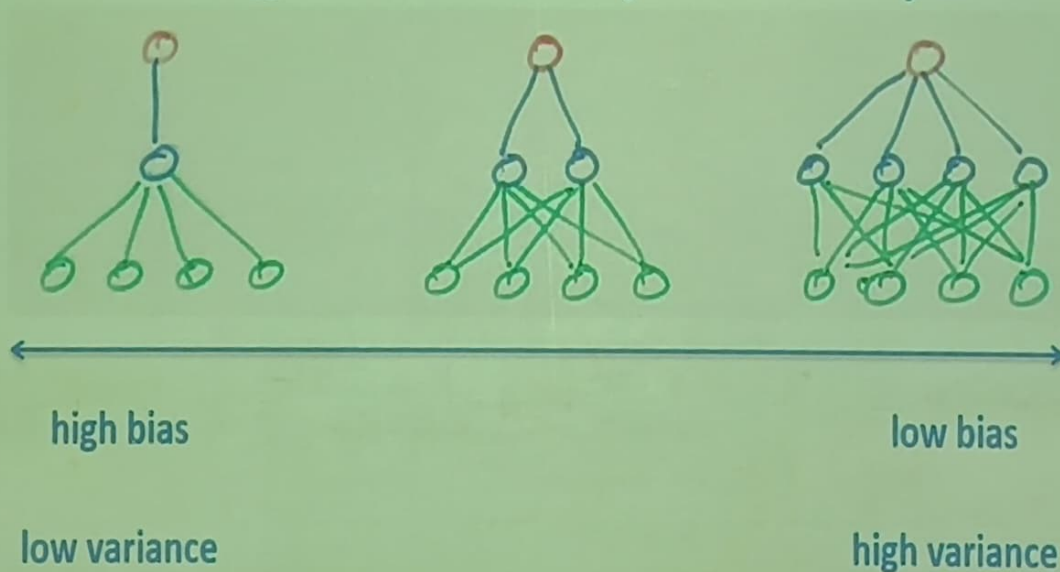


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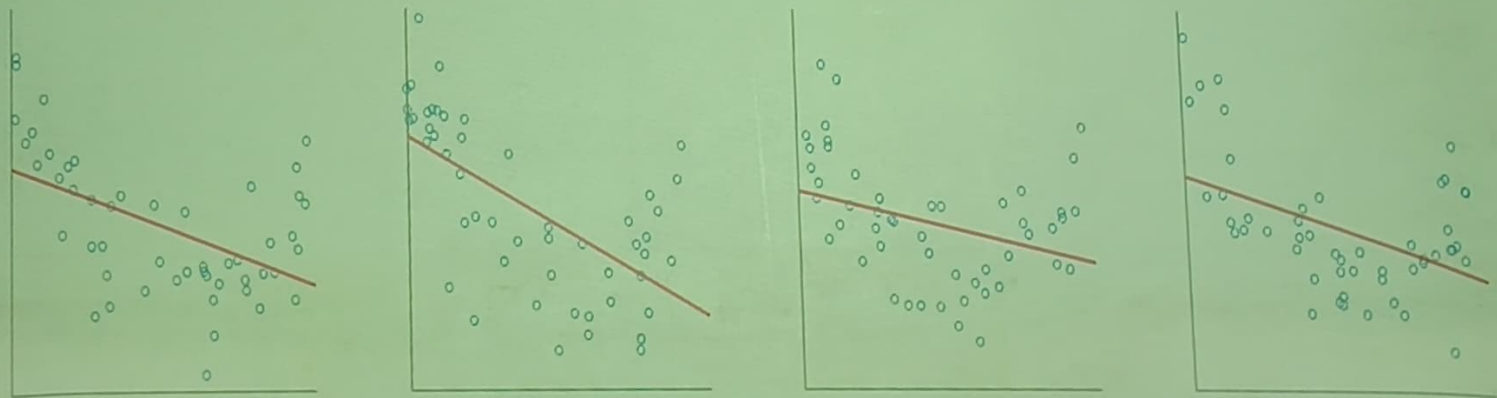
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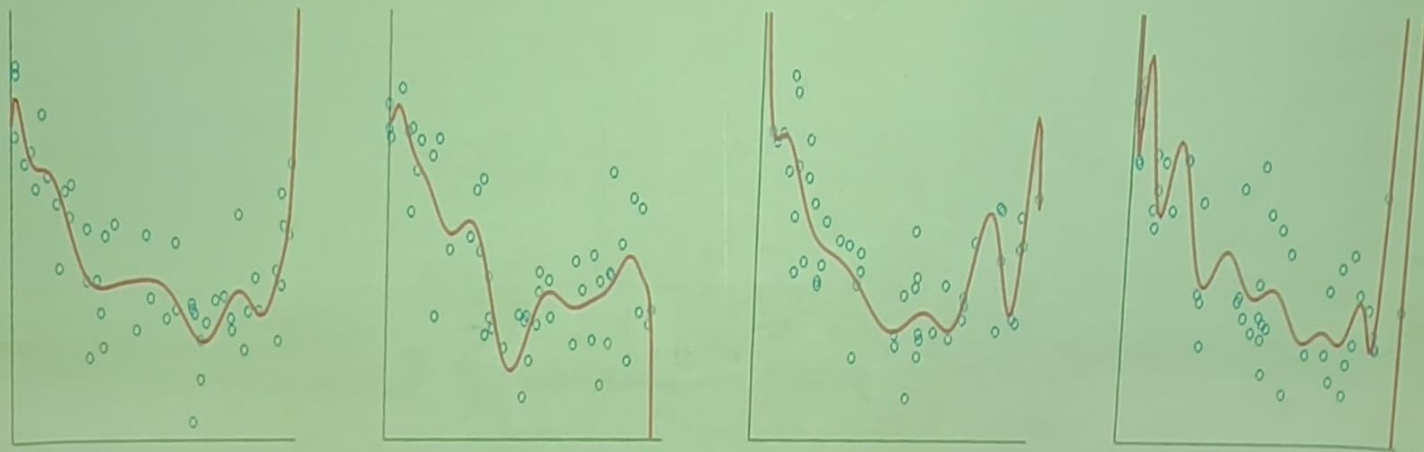


Bias



Regardless of training sample, or size of training sample, model will produce consistent errors

Variance



Different samples of training data yield different model fits

Formalizing Bias and Variance

Given data set

- $\mathcal{D} = \{(x_1, y_1), \dots, (x_N, y_N)\}$

And model built from data set,

- $f(x; \mathcal{D})$

We can evaluate the effectiveness of the model using mean squared error:

- $\text{MSE} = E \left[(y - f(x; \mathcal{D}))^2 \right]$

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- $\text{MSE} = E_{p(x,y,\mathcal{D})} \left[(y - f(x; \mathcal{D}))^2 \right]$

- with constant $|\mathcal{D}| = N$

$$\text{MSE}_x = E_{\mathcal{D}|x} \left[(y - f(x; \mathcal{D}))^2 \right]$$

$$\begin{aligned}
 \text{MSE}_x &= E_{\mathcal{D}|x} \left[(y - f(x; \mathcal{D}))^2 \right] \\
 &= (E_{\mathcal{D}}[f(x; \mathcal{D})] - E[y|x])^2 \\
 &\quad + E_{\mathcal{D}}[(f(x; \mathcal{D}) - E_{\mathcal{D}}[f(x; \mathcal{D})])^2] \\
 &\quad + E[(y - E[y|x])^2]
 \end{aligned}$$

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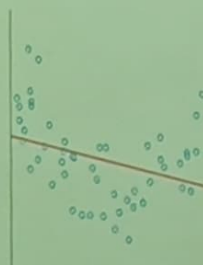
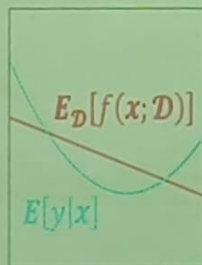
bias:

difference between
average model
prediction (across
data sets) and the
target

$$= (E_{\mathcal{D}}[f(x; \mathcal{D})] - E[y|x])^2$$

$$+ E_{\mathcal{D}}[(f(x; \mathcal{D}) - E_{\mathcal{D}}[f(x; \mathcal{D})])^2]$$

$$+ E[(y - E[y|x])^2]$$



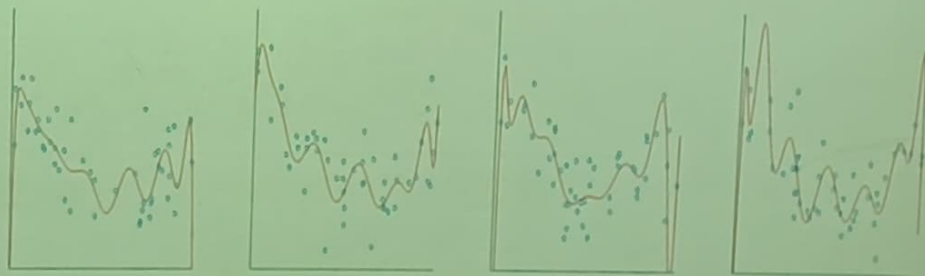
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variance of models
(across data sets)
for a given point

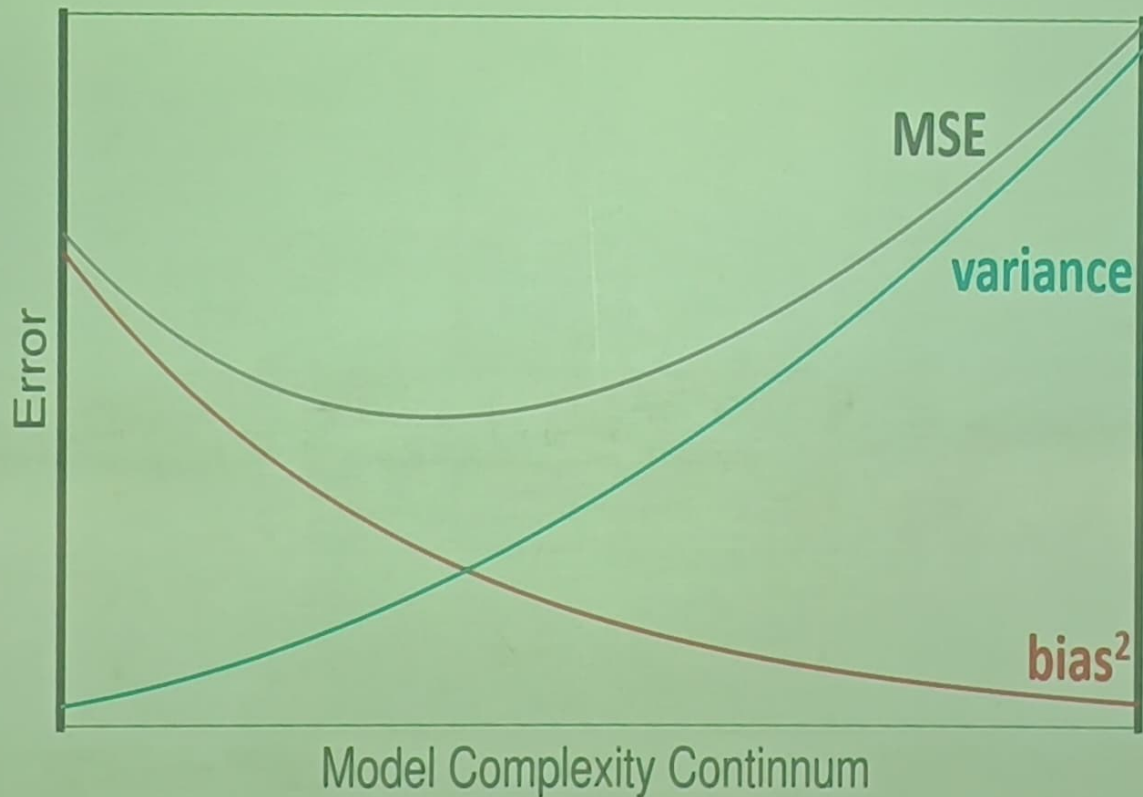


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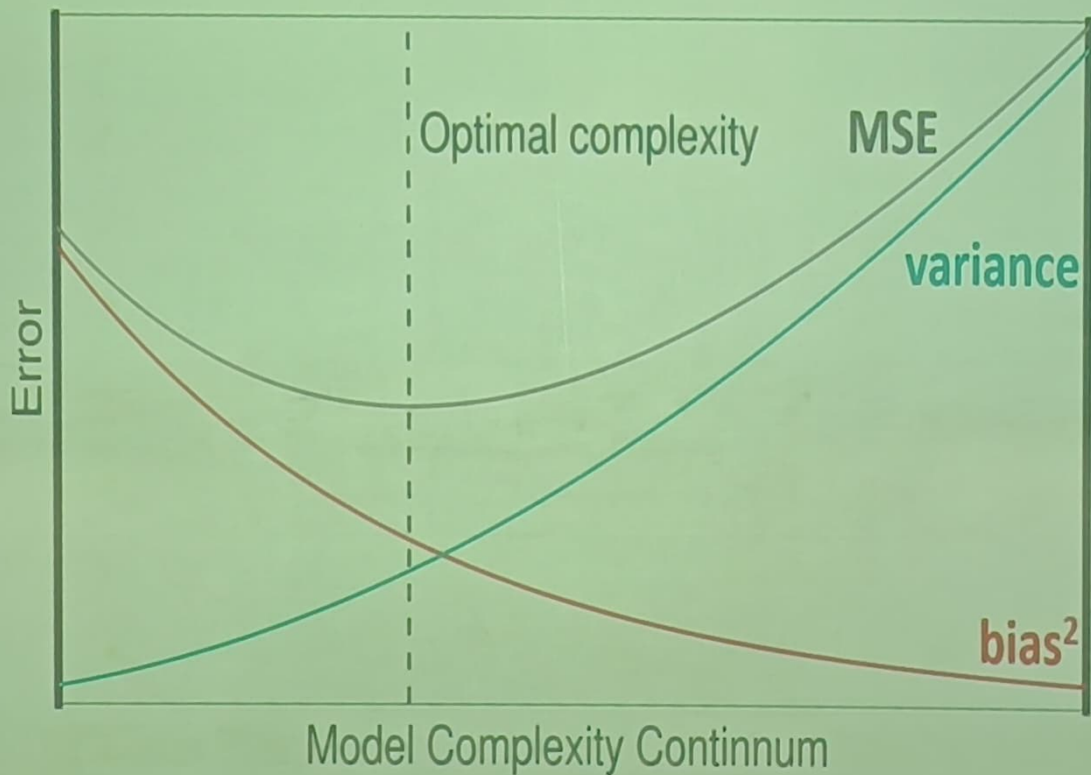
intrinsic noise in data set



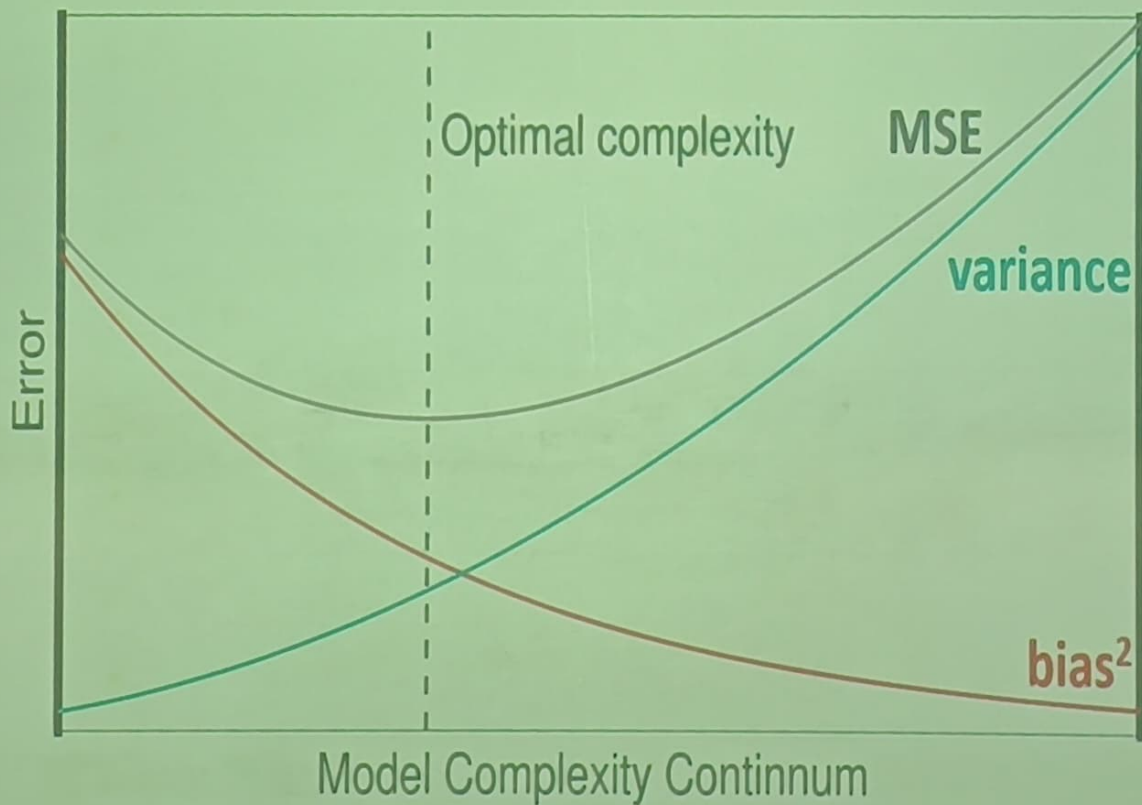
Bias-Variance Trade Off

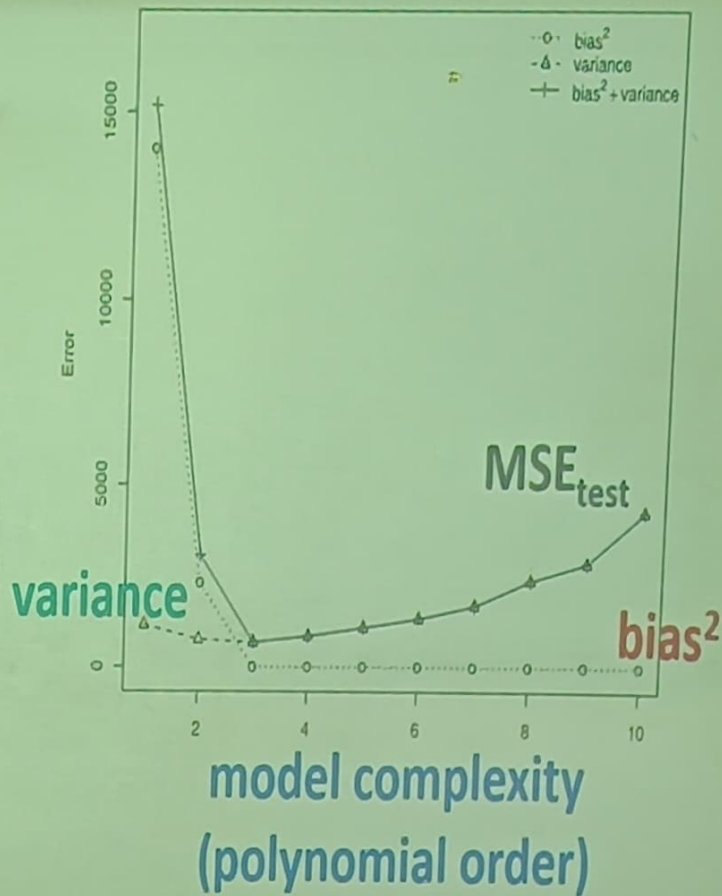
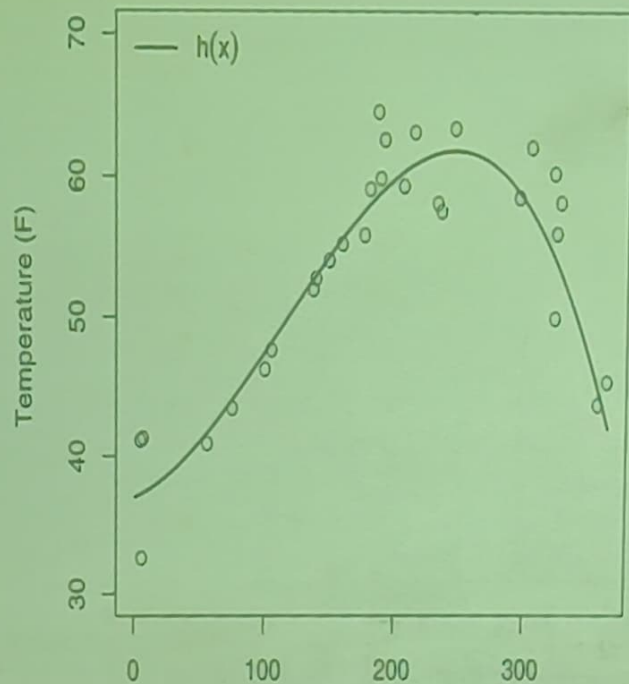


Bias-Variance Trade Off



Bias-Variance Trade Off

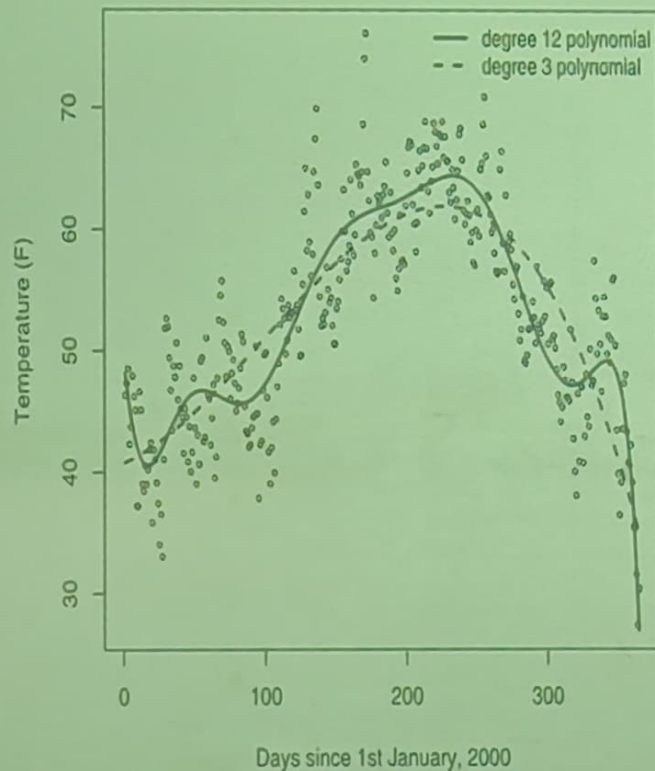




Gigerenzer & Brighton (2009)

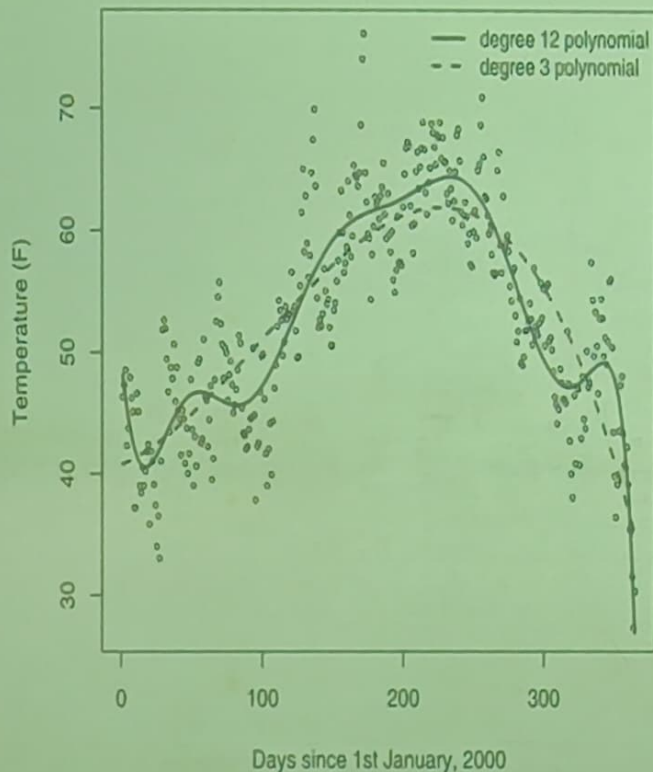
Bias-Variance Trade Off Is Revealed Via Test Set Not Training Set

London's daily temperature in 2000

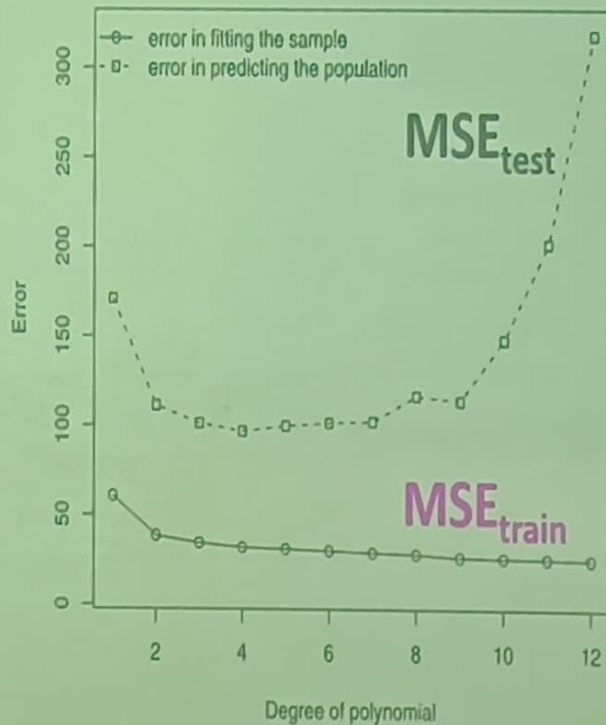


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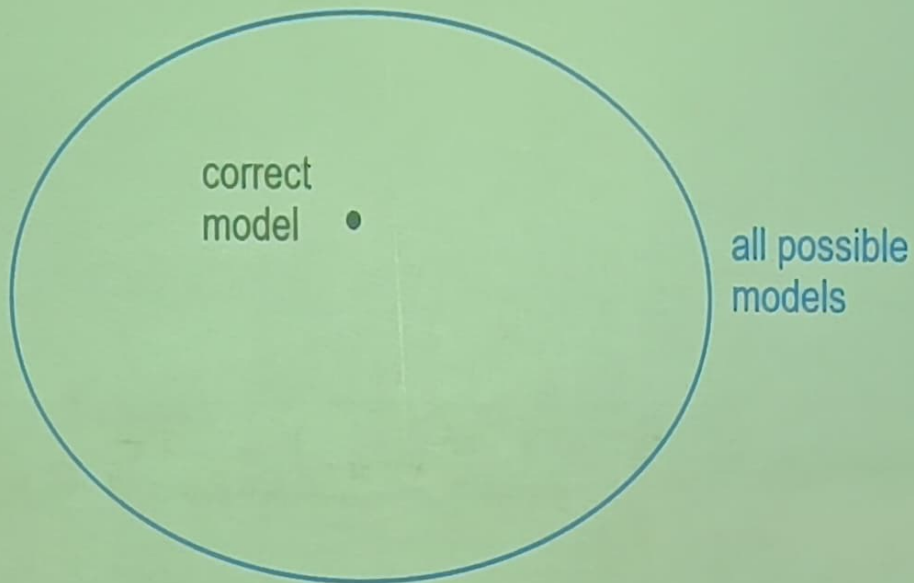
London's daily temperature in 2000



Model performance for London 2000 temperatures



Back To The Venn Diagram...



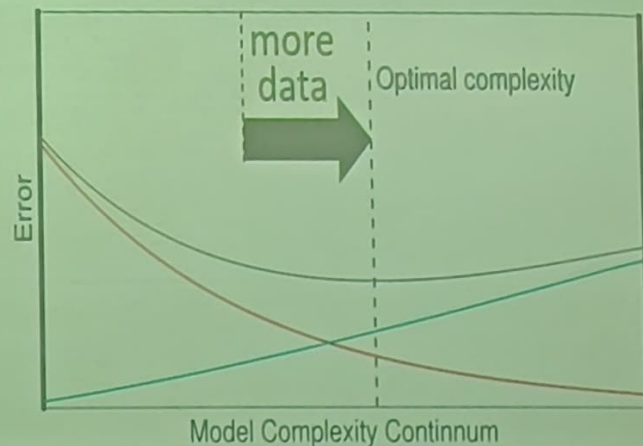
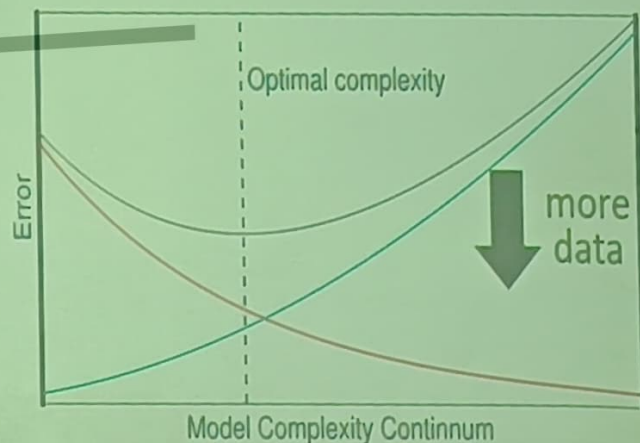
Current Perspective In Machine Learning

We can learn complex domains using

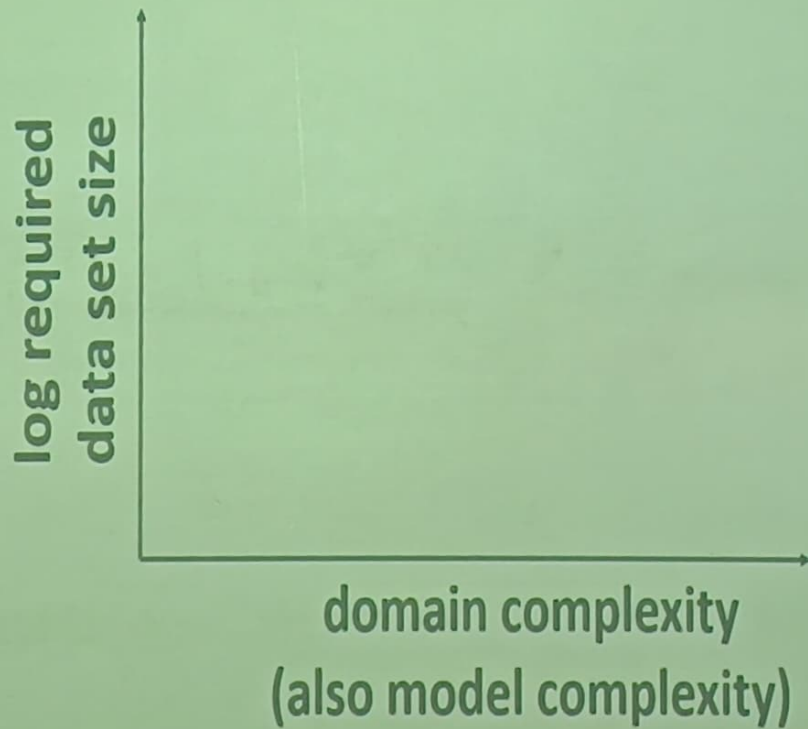
- low bias model (deep net)
- tons of training data

But will we have enough data?

- E.g., speech recognition



Scaling Function

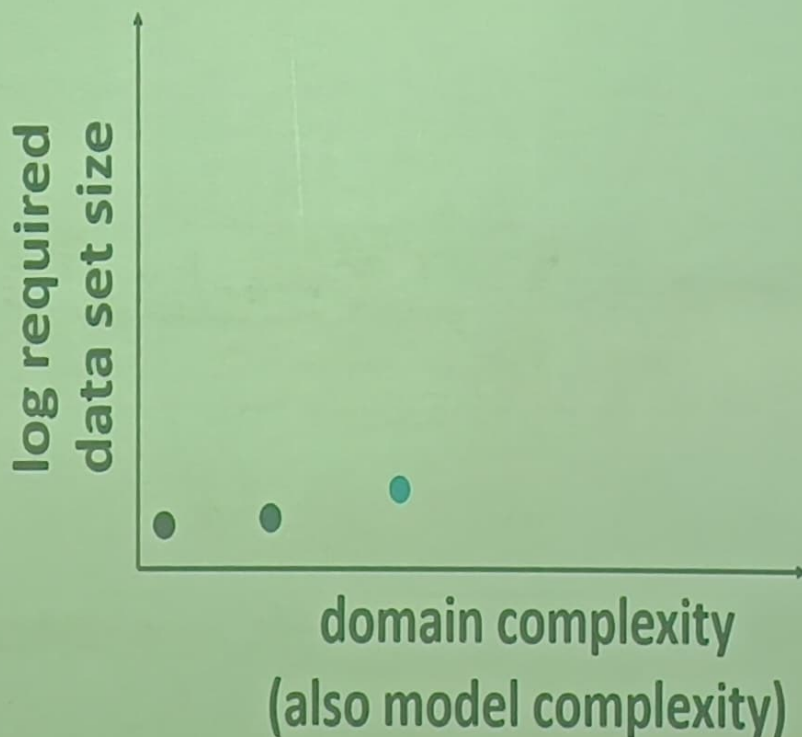


Scaling Function

Single-speaker,
small vocabulary,
isolated words

Multiple-speaker,
small vocabulary,
isolated words

Multiple-speaker,
small vocabulary,
connected speech



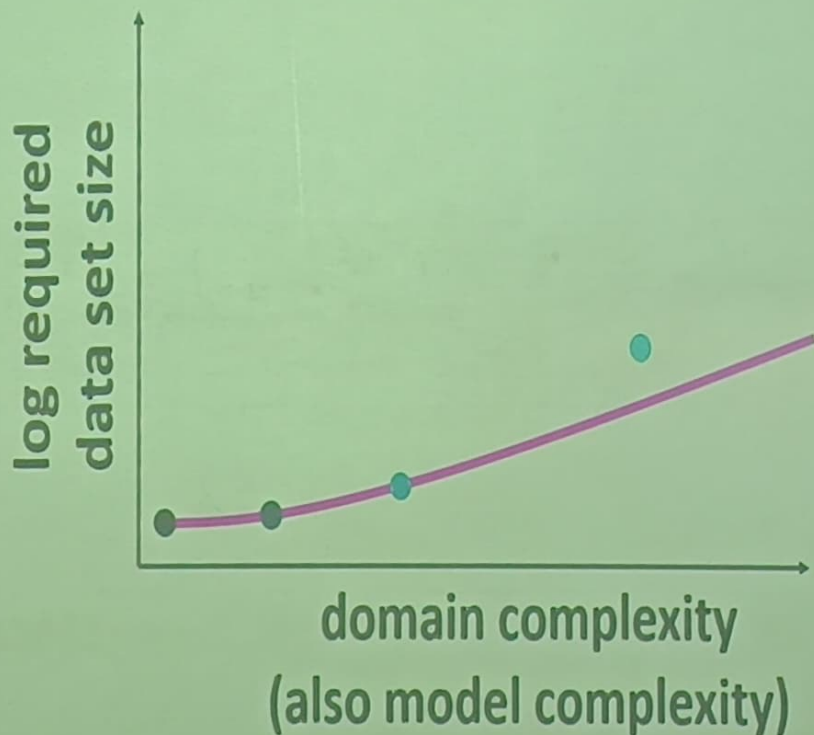
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Scaling Function

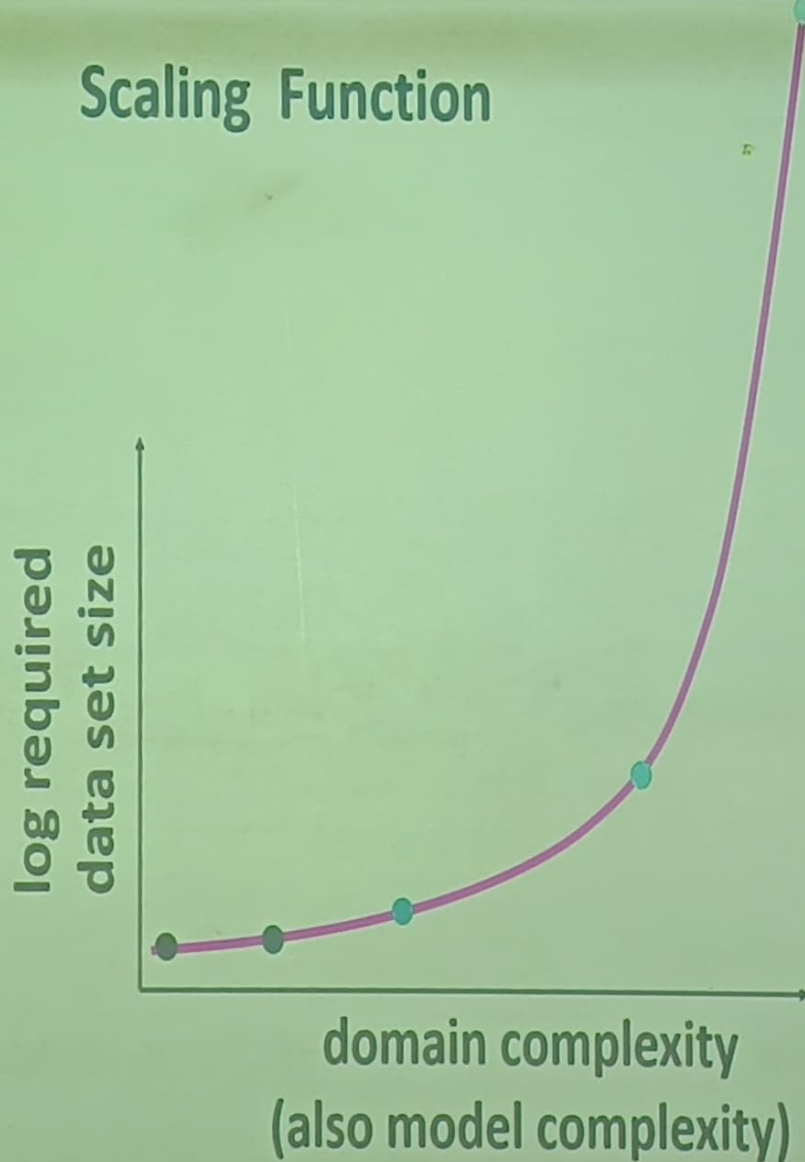
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Intelligent chatbot /
Turing test



The Challenge To AI

In the 1960s

- Neural nets (perceptrons) created wave of excitement
- But Minsky and Papert (1969) showed challenges to scaling



In the 1990s

- Neural nets (back propagation) created wave of excitement
- Worked great on toy problems but arguments about scaling (Elman et al., 1996; Marcus, 1998)



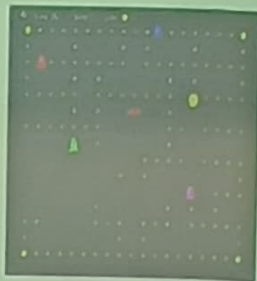
Now in the 2020s

- Neural nets (deep learning) created a wave of excitement
- Researchers have clearly moved beyond toy problems
- Nobody is yet complaining about scaling
- But there is no assurance that methods won't disappoint again

1995



2015



?

